

181. The number of incomplete orbitals in inner transition element is

- (a) 2
- (b) 3**
- (c) 4
- (d) 1

Answer: (b) 3

Explanation: The correct choice is 3. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

182. Most common oxidation states of Cs (cesium) are

- (a) + 2, + 3
- (b) + 2, + 4
- (c) + 3, + 4**
- (d) + 3, + 5

Answer: (c) + 3, + 4

Explanation: Assign oxidation numbers from charge balance and the accessible ns/(n-1)d valence electrons of the transition element.

183. The 3d elements show variable oxidation states because the energies of the following sets of orbitals are almost similar

- (a) ns,(n-1)d**
- (b) ns,nd
- (c) (n-1)s,nd
- (d) np,(n-1)d

Answer: (a) ns,(n-1)d

Explanation: Transition elements can use both ns and (n-1)d electrons in bonding because these levels are close in energy, so several oxidation states are possible.

184. Which of the following 3d bivalent metal ions has the smallest number of unpaired d electrons

- (a) 3d⁶
- (b) 3d⁷
- (c) 3d⁸
- (d) 3d⁹**

Answer: (d) 3d⁹

Explanation: The correct choice is 3d⁹. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

185. The 3d metal ions form coloured compounds because the energy corresponding to the following lies in the visible range of electromagnetic spectrum

- (a) Free energy change of complex formation by 3d metal ions
- (b) d-d transitions of 3d electrons**
- (c) Heat of hydration of 3d metal ions
- (d) Ionisation energy of 3d metal ions

Answer: (b) d-d transitions of 3d electrons

Explanation: Partly filled d orbitals allow electronic transitions that absorb visible light; d⁰ or d¹⁰ ions are generally



colourless unless charge-transfer effects operate.

186. The oxidation number of iron in potassium ferrocyanide is

- (a) +2
- (b) +3
- (c) +4
- (d) Zero

Answer: (a) +2

Explanation: Assign oxidation numbers from charge balance and the accessible $(n-1)d$ valence electrons of the transition element.

187. In transition elements, the orbitals partially filled by electrons are

- (a) s - orbitals
- (b) p - orbitals
- (c) d - orbitals
- (d) f - orbitals

Answer: (c) d - orbitals

Explanation: The correct choice is d - orbitals. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

188. Number of unpaired electrons in Mn^{2+} is

- (a) 3
- (b) 5

- (c) 4
- (d) 1

Answer: (b) 5

Explanation: Determine the d-electron count of the atom/ion and apply Hund's rule. More unpaired electrons give a larger spin-only magnetic moment; $d0$ and $d10$ species are diamagnetic.

189. Mercury is the only metal which is liquid at $0^\circ C$. This is due to its

- (a) Very high ionisation energy and weak metallic bond
- (b) Low ionisation potential
- (c) High atomic weight
- (d) High vapour pressure

Answer: (a) Very high ionisation energy and weak metallic bond

Explanation: An amalgam is an alloy/solution containing mercury; mercury has distinctive metallic bonding and amalgamation behaviour.

190. Essential constituent of an amalgam is

- (a) Iron
- (b) An alkali metal
- (c) Silver
- (d) Mercury

Answer: (d) Mercury

Explanation: An amalgam is an alloy/solution containing mercury; mercury has distinctive metallic bonding and amalgamation behaviour.



191. Mercury is transported in metal containers made of

- (a) Silver
- (b) Lead
- (c) Iron**
- (d) Aluminium

Answer: (c) Iron

Explanation: An amalgam is an alloy/solution containing mercury; mercury has distinctive metallic bonding and amalgamation behaviour.

192. The electroplating of chromium is undertaken because

- (a) Electrolysis of chromium is easier
- (b) Chromium can form alloys with other metals
- (c) Chromium gives protective and decorative coating to the base metal**
- (d) Of the high reactivity of metallic chromium

Answer: (c) Chromium gives protective and decorative coating to the base metal

Explanation: The correct choice is Chromium gives protective and decorative coating to the base metal. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

193. An element having the electronic configuration $Ar3d^24s^2$ belongs to

- (a) s - block elements
- (b) p - block elements
- (c) d - block elements**
- (d) f - block elements

Answer: (c) d - block elements

Explanation: Use the Aufbau order with the known transition-metal exceptions; when forming cations, ns electrons are removed before (n-1)d electrons.

194. Which one of the following is not a transition metal

- (a) Chromium
- (b) Titanium
- (c) Lead**
- (d) Tungsten

Answer: (c) Lead

Explanation: The correct choice is Lead. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

195. The atomic number of an element is 22. The highest oxidation state exhibited by it in its compounds is

- (a) 1
- (b) 2
- (c) 3
- (d) 4**

Answer: (d) 4



Explanation: Assign oxidation numbers from charge balance and the accessible $ns/(n-1)d$ valence electrons of the transition element.

196. d-block elements form

- (a) Ionic compounds
- (b) Covalent compounds
- (c) Ionic and covalent compounds**
- (d) Only complex compounds

Answer: (c) Ionic and covalent compounds

Explanation: The correct choice is Ionic and covalent compounds. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

197. The transition metals have a less tendency to form ions due to

- (a) High ionisation energy
- (b) Low heat of hydration of ion
- (c) High heat of sublimation
- (d) All of these**

Answer: (d) All of these

Explanation: The correct choice is All of these. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

198. The electronic configuration of Ag atom is

- (a) $Kr3d104s1$
- (b) $Xe4f14d106s1$
- (c) $Kr4d105s1$**
- (d) $Kr4d95s2$

Answer: (c) $Kr4d105s1$

Explanation: Use the Aufbau order with the known transition-metal exceptions; when forming cations, ns electrons are removed before $(n-1)d$ electrons.

199. Most powerful oxidizing property of manganese is shown by which of the following oxidation state

- (a) $Mn(+7)$**
- (b) $Mn(+2)$
- (c) $Mn+4$
- (d) $Mn(+5)$

Answer: (a) $Mn(+7)$

Explanation: Assign oxidation numbers from charge balance and the accessible $ns/(n-1)d$ valence electrons of the transition element.

200. Which one of the following ions is colourless

- (a) $Cu+$**
- (b) Co^{2+}
- (c) Ni^{2+}
- (d) Fe^{3+}

Answer: (a) $Cu+$





Explanation: Partly filled d orbitals allow electronic transitions that absorb visible light; d0 or d10 ions are generally colourless unless charge-transfer effects operate.

